

Program 8: Create Kubernetes Pods using YAML Descriptors

Project Structure

```
program8
|--nodejs-configmap.yaml
|--nodejs-pod.yaml
|--server.js
```

Start the minikube

```
minikube start --driver=docker
```

Step1:

Create a node js app

Nano server.js

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program8$ cat server.js
const http = require("http")
const port = 3000

const server=http.createServer((req,res)=>{
res.end("Hello from Nodejs inside kubernetes!");
});

server.listen(port,()=>{
console.log('Server running at port ${port}');
});
```

Step2: create a nodejs-configmap.yaml file

```
Normal Started 27M kubernetes Started container nodejs
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program8$ cat nodejs-configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require("http");
    const port = 3000;
    const server = http.createServer((req, res) => {
      res.end("Hello from Node.js inside Kubernetes!");
    });

    server.listen(port, () => {
      console.log(`Server running at port ${port}`);
    });
```

Step 3: create a nodejs-pods.yaml file

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program8$ cat nodejs-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: program-8
  labels:
    app: nodejs
spec:
  containers:
    - name: nodejs
      image: node:18
      command: ["node", "/usr/src/app/server.js"]
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app
      ports:
        - containerPort: 3000
  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config
```

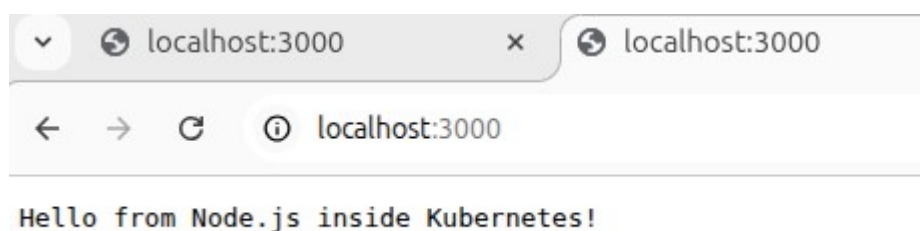
Step4: Apply the yaml files

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program8$ kubectl apply -f nodejs-configmap.yaml
configmap/nodejs-app-config unchanged
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program8$ kubectl apply -f nodejs-pod.yaml
pod/program-8 created
```

Step6: forward the port and access the nodejs app

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program8$ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
program-8     1/1     Running   0           6m38s
```

Step7: check the output on <http://localhost:3000>



```
bash: syntax error near unexpected token `}'
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program8$ kubectl delete pods --all
pod "program-8" deleted
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program8$ kubectl delete svc --all
service "kubernetes" deleted
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program8$ kubectl delete deployments --all
No resources found
```